

Database Design – Justification of Design Decisions

The MoMo SMS Analyzer database is structured around seven tables in Third Normal Form (3NF), ensuring every attribute depends solely on its primary key and eliminating redundancy across the schema.

The **transactions** table serves as the central entity, linked to **users** via two separate nullable foreign keys — **sender_id** and **receiver_id** — both referencing the same table. This self-referencing approach handles three real-world scenarios: internal-to-internal transfers, external sender to internal receiver (e.g. bank deposits), and internal sender to external receiver (e.g. merchant payments), without duplicating the users table.

transaction_categories was separated into its own lookup table rather than hardcoded as an ENUM on transactions. This allows new categories to be added without schema migrations and supports attaching metadata such as the **is_income** flag used in analytics aggregations.

The **tags** and **transaction_tags** tables resolve a genuine many-to-many relationship — a transaction can carry multiple labels, and a label can apply across many transactions. The junction table uses a composite primary key (**transaction_id**, **tag_id**) to prevent duplicate assignments, with CASCADE deletes to automatically clean orphaned links.

uploaded_files is linked to transactions to maintain full traceability from raw XML source to processed record, supporting re-processing workflows if a parsing bug is discovered.

system_logs uses a nullable foreign key to transactions, allowing the audit trail to capture file-level and user-level events that are not tied to a single transaction record.

Finally, ENUM types on **transaction_type**, **transaction_status**, and log **status** enforce domain integrity at the engine level, providing a safety net independent of application code.